

(KTXFI2EBNF) Physics II. Exam 2
January 5, 2026

I. Moving Reference Frames. Special Theory of Relativity

1. State the Galilean principle of relativity. (5 points)
2. Draw the experimental setup of the Michelson–Morley interferometer, indicate and name the instruments involved in the experiment. What was the objective of performing this experiment? What was the outcome of the experiment? (5 points)

II. Limits of the Classical Conceptual Framework

3. What are the Stefan–Boltzmann law and Wien’s displacement law? Describe them in your own words and also give their mathematical expressions. Specify the physical meaning of the symbols appearing in the formulas. (5 points)
4. What is the Compton effect, what did it prove, and how was this demonstrated? (5 points)
5. Describe Planck’s hypotheses; that is, explain the significance of Planck’s radiation law. (5 points)

III. Elements of Quantum Mechanics. Physics of Condensed Matter. Theory of Electrical Conduction

6. Describe the electrical conduction mechanism of semiconductors using the band theory of solids. What characterizes semiconductors from the point of view of electrical conduction? (5 points)
7. What is the tunneling effect? Describe the phenomenon in your own words. (5 points)

IV. Development of Atomic Theory

8. Describe the Rutherford scattering experiment. (5 points)
9. What is the Zeeman effect? Describe the phenomenon. Under what conditions does it occur and how does it manifest itself? (5 points)
10. Given the element with atomic number 11 in the periodic table, write down its electronic configuration using the standard atomic physics notation. (5 points)

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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